Sigmoid: An "s" shaped curve used in neural networks.

Signal: A description of how one parameter varies with another parameter. Example: a *voltage* that varies with *time*.

Signal restoration: Returning a signal to its original form after it has been changed or degraded in some way. One of the main uses of *filtering*.

Sinc function: Formally defined by the relation: $sinc(a) = sin(\pi a)/\pi a$. The π terms are often hidden in other variables, making it in the general form: $sin(x)/x$. Important because it is the Fourier transform of the rectangular pulse.

Single precision: A floating point notation that used 32 bits to represent each number. See *double precision* for comparison.

Single-pole digital filters: Simple recursive filters that mimic *RC* high-pass and low-pass filters in electronics.

Sinusoidal fidelity: An important property of linear systems. A sinusoidal input can only produce a sinusoidal output; the amplitude and phase may change, but the frequency will remain the same.

Sonar: Sound Navigation And Ranging. The use of sound to detect submarines and other underwater objects. *Active sonar* uses echo location, while *passive sonar* only listens.

Source code: A computer program in the form written by the programmer; distinguished from *executable code*, a form that can be directly run on a computer.

Spatial domain: A signal having distance (space) as the independent variable. Images are signals in the spatial domain.

Spectral analysis: Understanding a signal by examining the amplitude, frequency, and phase of its component sinusoids. The primary tool of spectral analysis is the Fourier transform.

Spectral inversion: Method of changing a filter kernel such that the corresponding frequency response is flipped top-for-bottom. This can change low-pass filters to high-pass, band-pass to band-reject, etc.

Spectral leakage: Term used in spectral analysis. Since the DFT can only be taken of a finite length signal, the frequency spectrum of a sinusoid is a peak with tails. These tails are referred to as *leakage* from the main peak.

Spectral reversal: Technique for changing a filter kernel such that the corresponding frequency response is flipped left-for-right. This changes low-pass filters into high-pass filters.

Spectrogram: Measurement of how an audio frequency spectrum changes over time. Usually displayed as an image. Also called a *voiceprint*.

Standard deviation: A way of expressing the fluctuation of a signal around its average value. Defined as the square-root of the average of the deviations squared, where the deviation is the difference between a sample and the mean. See *root-mean-square* for comparison.

Static linearity: Refers to how a linear system acts when the signals are not changing (i.e., they are *DC* or *static*). In this case, the output is equal to the input multiplied by a constant.

Statistical noise: Variations in a signal's value resulting from it being represented by a finite number of particles, such as: x-rays, electrons, or light photons. Also called *Poisson statistics* and *Poisson noise*.

Steepest descent: Strategy used in designing iterative algorithms. Analogous to finding the bottom of a valley by always moving in the down-hill direction.

Step response: The output of a system when the input is a step function.

Stopband: The band of frequencies that a filter is designed to block.

Stopband attenuation: The amount by which frequencies in the stopband are reduced in amplitude, usually expressed in decibels. Used to describe a filter's performance.

Substitution: A way of using complex numbers to represent a physical problem, such as electric circuit design. In this method, *j* terms are added to change the physical problem to a complex form, and then removed to move back again. See *mathematical equivalence* for comparison.

Switched capacitor filter: Analog filter that uses rapid switching to replace resistors. Made as easy-to-use integrated circuits. Often used as antialias filters for ADC and reconstruction filters for DAC.